

Figure 1. Decode throughput after each fused kernel.

NVIDIA L40S, selective INT4 weights, single stream, greedy, 17-token prompt, 64 generated tokens, inside a hard 32 GiB process cap. Bars are the median of five repeats in one container; the interval on the final bar is the range across five independent containers, 113.77 to 114.03. The kernels are not bit-identical to the reference path: teacher forced they agree on 96.9 percent of next-token choices at 0.0036 nats mean KL, roughly one tenth of the divergence INT4 quantisation itself introduces. Source: [engine/kernels/RESULTS.md](#)

